

# On Robust 3D Human Pose Estimation and Forecasting

Sazan Mahbub   Chuong Huynh   Jayesh Jayashankar   Aswath Muthuselvam  
University of Maryland, College Park

{smahbub, chuonghm, jayeshj, aswath}@umd.edu

## Abstract

*Current human pose estimation and forecasting models perform very well with video inputs. However, those models will achieve better results if domains and gradient of loss functions are taken into account. In this work, we present the new OHJM+WAO losses for the human pose estimation and the DA-STS-GCN framework for the human pose forecasting. All modifications produce more accurate results than the state-of-the-art baselines by a considerable margin. While OHJM+WAO can decrease the error by 0.1-0.6 mm, the average MPJPE drops by around 3 mm with DA-STS-GCN on our benchmark.*<sup>1</sup>

## 1. Introduction

Human pose estimation (HPE) is an essential task in computer vision. The human pose is a skeleton graph with nodes or joints representing the body’s parts and edges representing their connections. HPE requires localizing joints from input data such as images and videos. Human pose forecasting (HPF) is built on HPE to predict future human poses. While HPE can be applied on static images, videos are typical HPF inputs because of the requirement of previous human poses. The two tasks are directly relevant when HPE output can be HPF input, and HPF’s prediction can be prior to HPE. Depending on the space dimension and camera views, there are different HPE/HPF problems variants. This project looks at 3D HPE/HPF from single view inputs.

In HPE, a straightforward method [18, 19] is to estimate 3D human pose directly from images or video frames. Adding a middle stage of predicting 2D poses before lifting to 3D poses recently produced many improvements [2, 13, 35]. To reduce errors propagating from images, one robust solution [20, 38] is to use video temporal information to capture joint global motions between frames. Modern 2D-to-3D lifting HPE relies on deep learning [26, 30], especially models for sequential data [1, 10], especially Transformer [27]. PoseFormer [36] is a transformer network designed for 2D-to-3D HPE. This transformer model for 3D

HPE produces more accurate estimated 3D poses than all previous works. In our work, we got more accurate results with this baseline by tweaking the loss function.

In terms of HPF, previous methods [7, 16, 25, 29] leveraged Graph Convolutional Networks [9], Recurrent Neural Networks [5], and Transformers [27] to encode spatial and temporal interaction. However, these HPF methods can potentially run into the drawback of overfitting on the human subjects and lose the ability to generalize for pose forecasting. To side-step this issue we propose *Domain Adaptive STS-GCN* (DA-STS-GCN), a method that leverages domain adaptive adversarial learning using the reverse gradient approach with the SOTA method. To our best knowledge, this is the first-ever successful application of domain adaptation for HPF problems. We perform experiments to compare against previous SOTA methods and carry out ablation studies to verify the domain adaptive nature of the model in various experimental scenarios.

## 2. Related Works

**Joint angle awareness in HPE.** Numerous joint position and angle loss functions have been proposed in the HPE literary works. Constraints on the bone length and joint angle orientation [31] improve position estimation accuracy. As proposed in [37], a joint and position smoothness function reduces the magnitude of loss as position errors approach ground truth. Additionally, the projection difference between successively connected limbs is considered a loss function. By analyzing the limit of angles between joints and planes, the advanced angle constraints proposed in [17] demonstrated the effectiveness of angles in HPE.

**Gradient clipping.** There are always certain items that the model finds challenging to train. With the idea of OHEM [24], hard samples with false positives receive more consideration during training. In contrast, Wing loss [4] reduces the harmful effects of outliers in training.

**Motion HPF.** When it comes to forecasting, there are a few approaches which use a generative 3D human motion prior to model the next future frame with a conditional variational autoencoder (CVAE) [12, 22]. On the other hand, for forecasting longer sequence (our focus in this paper), several

<sup>1</sup>Code is publicly available at <https://github.com/hmchuong/3DHPE-HPF>

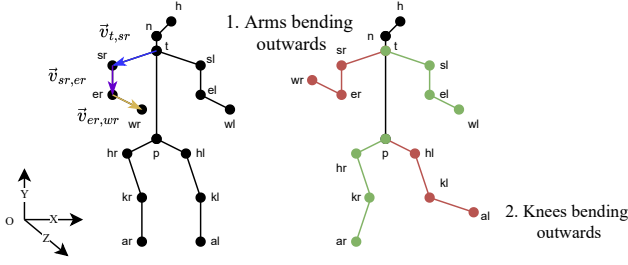


Figure 1. The skeleton defined in the Human3.6M dataset and two illegal cases at elbows and knees. There are 17 joints in 3D space defining the skeleton. The highlighted limbs in red are illegal cases when the  $\vec{v}_{er,wr}$  has opposite direction to  $\vec{O}z$  and  $\vec{v}_{kl,al}$  is in the same direction to  $\vec{O}z$

previous methods encode *space* and *time* both within a single module [7, 16, 29], and separately *et al.* [25].

**Adversarial Domain Adaptation.** Domain adaptation is vital for any method to improve its robustness across different distributions. [34] introduces the gradient reversal layer by promoting the emergence of features that are both discriminative and domain-invariant. [21] proposed a siamese architecture for adversarial learning by combining natural and synthetic images into a single domain when training networks. [32] uses a triplet loss for adversarial learning to eliminate effects of domain shifts with a domain similarity discriminator.

### 3. Methodology

#### 3.1. Angle awareness and gradient clipping for HPE

##### 3.1.1 Joint angle losses

Fig. 1 represents the 17-joint skeleton that is used in the Human3.6M dataset [6]. Based on this skeleton, several angle constraints are investigated in this project. Firstly, the limb angle loss (AngLoss) in [37] compares the difference between estimated and groundtruth human pose in eight angles at shoulders ( $sr, sl$ ), ankles ( $ar, al$ ), hips ( $hr, hl$ ) and knees ( $kr, kl$ ). Secondly, the limb and torso angle orientation constraints (AngOrient) were proposed in [31]. Lastly, we tested the performance of the advanced angle constraints (AngLimit) [17]. This loss function requires the analysis of lower and upper bounds of angles between complex planes and vectors in the human skeleton.

As shown later in Sec. 4.1.1, the AngOrient is the most effective loss function in our experiments. In the limb orientation loss, the direction of vectors  $\vec{v}_{er,wr}$ ,  $\vec{v}_{el,wl}$ ,  $\vec{v}_{kr,ar}$  and  $\vec{v}_{kl,al}$  are constrained by adjacent vectors. The two illegal and legal cases are highlighted in red and green respectively in Fig. 1. Particularly, the limb orientation loss  $\mathcal{L}_{LO}$  is defined as:

$$\begin{aligned} \mathcal{L}_{LO} = & \max(-\vec{v}_{t,sr} \times \vec{v}_{sr,er} \cdot \vec{v}_{er,wr}, 0) \\ & + \max(-\vec{v}_{sl,el} \times \vec{v}_{t,sl} \cdot \vec{v}_{el,wl}, 0) \\ & + \max(-\vec{v}_{hr,kr} \times \vec{v}_{p,hr} \cdot \vec{v}_{kr,ar}, 0) \\ & + \max(-\vec{v}_{p,hl} \times \vec{v}_{hl,kl} \cdot \vec{v}_{kl,al}, 0) \end{aligned} \quad (1)$$

The torso orientation constraints reflect that some vectors have opposite directions in the skeleton. The torso orientation loss  $\mathcal{L}_{TO}$  is defined as:

$$\mathcal{L}_{TO} = \max(\vec{v}_{n,h} \cdot \vec{v}_{t,p}, 0) + \max(\vec{v}_{t,sr} \cdot \vec{v}_{t,sl}, 0) + \max(\vec{v}_{p,hr} \cdot \vec{v}_{p,hl}, 0) \quad (2)$$

##### 3.1.2 Wing Angle Orientation

With joint error loss, only one global optimal solution exists when the keypoints between groundtruth and estimation are coincident. With angle losses, the angles do not change when moving keypoints along the edges. Therefore, angle loss functions cause unstable gradients during training. In other words, angle loss functions are very sensitive to outliers. To reduce the sensitivity of angle losses, we apply the wing( $x$ ) function from Wing loss [4]. In our experiment, we choose  $w = 5 \times 10^{-3}$  and  $\epsilon = 10^{-3}$ .

##### 3.1.3 Online Hard Joint Mining

Mean Per Joint Position Error (MPJPE), which is a standard metric in HPE and HPF, is used as a loss function in many HPE papers. Let  $Y, \hat{Y} \in \mathbb{R}^{N \times J \times 3}$  denote the groundtruth and prediction of the 3D human pose and  $N, J$  are the number of samples and joints. The MPJPE is the average of  $L_2$  distance between each joint in  $Y$  and  $\hat{Y}$ :

$$\mathcal{L}_{MPJPE} = \frac{1}{N} \frac{1}{J} \sum_{i=1}^N \sum_{k=1}^J \text{dist}_{i,k} \quad (3)$$

where  $\text{dist}_{i,k} = \|Y_k^i - \hat{Y}_k^i\|_2$  and  $Y_k^i, \hat{Y}_k^i \in \mathbb{R}^3$  are coordinates of joints  $k$  at  $i$ -th pose in groundtruth and prediction respectively.

In our experiment, MPJPE becomes significantly small after several training epochs, which leads to no parameter updates. Inspired by OHEM [24], the loss value is computed as the average value of top  $m \in [0, 1]$  proportion of joints with highest errors. The algorithm proceeds as follows. Let  $\Delta \in \mathbb{R}^{N \times J}$  represent the distance matrix such that each entry at  $i$ -th row and  $k$ -th column is the distance  $\text{dist}_{i,k}$ . After flattening  $\Delta$  to vector  $\delta \in \mathbb{R}^{(N \cdot J) \times 1}$ , we sort all values. By taking  $s = \lfloor m \times N \times J \rfloor$  highest values in  $\delta$ , a new vector  $\delta'$  contains all errors of hard examples in the training set. The Online Hard Joint Mining (OHJM) is computed as  $\mathcal{L}_{OHJM} = \frac{1}{s} \sum_{i=1}^s \delta'_i$  where  $\delta'_i$  is the value of entry  $i$  in vector  $\delta'$ . In our experiment, we choose  $m = 0.7$  for the best result.

##### 3.1.4 Overall loss function

The final loss function is the weighted sum between  $\mathcal{L}_{OHJM}$ ,  $\mathcal{L}_{LO}$ , and  $\mathcal{L}_{TO}$  where  $\mathcal{L}_{LO}$  and  $\mathcal{L}_{TO}$  is smoothed by the wing function:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{OHJM} + \lambda_2 \text{wing}(\mathcal{L}_{TO}) + \lambda_3 \text{wing}(\mathcal{L}_{LO}) \quad (4)$$

where  $\lambda_1, \lambda_2, \lambda_3$  are balanced weights. We choose  $\lambda_1 = 0.6, \lambda_2 = \lambda_3 = 0.5$  in our experiments.

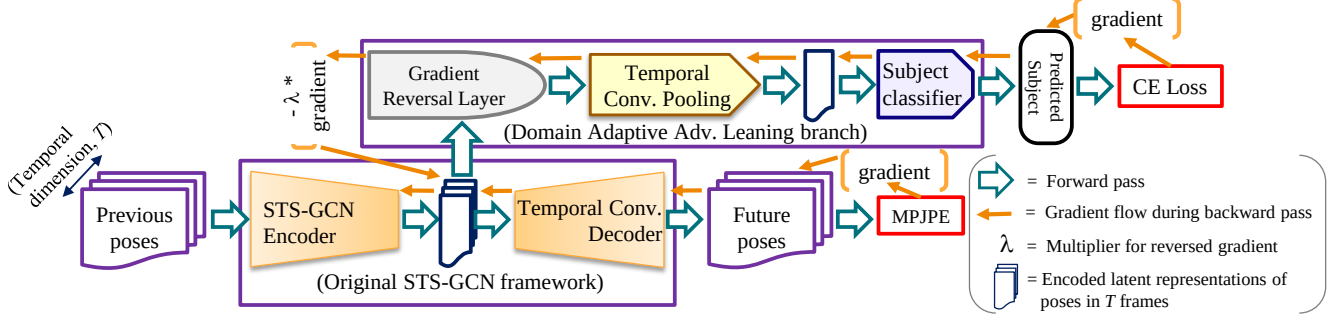


Figure 2. **The overall architecture of our proposed Domain Adaptive STS-GCN framework.** Here the input is a series of poses of a human subject in a video (i.e., previous poses). It passes through the *STS-GCN framework* to generate a series of consecutive poses in the future. On the other hand, in the upper branch (*Domain Adaptive Adversarial Learning branch*), the encoded hidden represented generated by the STS-GCN Encoder is used to predict the human subject present in the video.

### 3.2. Domain Adaptive STS-GCN for HPF

Let  $\mathbf{X}_{in} = [x_1, x_2, \dots, x_{\mathcal{T}}]$  be the 3D human pose history including  $\mathcal{T}$  poses where each pose  $x_i \in \mathbb{R}^{\mathcal{J} \times 3}$  where  $\mathcal{J}$  is the number of joints. Given  $\mathbf{X}_{in}$ , the HPF models need to output the future  $\mathcal{F}$  poses  $\mathbf{X}_{out} = [x_{\mathcal{T}+1}, x_{\mathcal{T}+2}, \dots, x_{\mathcal{T}+\mathcal{F}}]$ . In this section, we will describe the modification on state-of-the-art HPF baseline, particularly domain adaptive adversarial learning which can leverage information from subjects/actors to improve performance itself.

#### 3.2.1 Spatial and Temporal Graph Convolution

Previous state-of-the-art methods [11, 15, 16, 25], leverage graph representation to encode the spatial (within each frame) and the temporal (across frames) interaction in  $\mathbf{X}_{in}$  using graph convolutional network (GCN) [9], where  $\mathbf{X}_{in}$  is represented as a graph  $\mathcal{G}$ . The set of vertices  $\mathcal{V}(\mathcal{G})$  ( $|\mathcal{V}(\mathcal{G})| = \mathcal{T}\mathcal{J}$ ) in  $\mathcal{G}$  is the set of body joints of human  $\mathcal{H}$  in all  $\mathcal{T}$  frames in the video. The interaction between the vertices are encoded by edges, represented by adjacency matrices. These adjacency matrices can be predefined [9, 28] or can be learned along with the rest of the parameters in a graph neural network [25, 33]. Since, the latter approach was empirically shown to be promising for pose forecasting, we leverage this approach in our study. For Spatial Temporal GCN (ST-GCN) [25, 33], a layer  $l$  encodes information following Equation 5.

$$h^{(l+1)} = \Omega(\mathcal{A}_{st}^{(l)} h^{(l)} \mathcal{W}_{gcn}^{(l)}) \quad (5)$$

Here,  $h^{(l)} \in \mathbb{R}^{C^{(l)} \times \mathcal{J} \times \mathcal{T}}$  and  $h^{(l+1)} \in \mathbb{R}^{C^{(l+1)} \times \mathcal{J} \times \mathcal{T}}$  are feature representations of the vertices before and after applying graph convolution (where  $C^{(l)}$  and  $C^{(l+1)}$  are dimensions of these feature representations respectively) and  $\Omega(\cdot)$  is a non-linearity function. This equation represents a learnable function, parameterized by weights  $\mathcal{W}_{gcn}^{(l)} \in \mathbb{R}^{C^{(l)} \times C^{(l+1)}}$  and adjacency matrix  $\mathcal{A}_{st}^{(l)} \in \mathbb{R}^{\mathcal{T}\mathcal{J} \times \mathcal{T}\mathcal{J}}$ .

On the other hand, [25], empirically showed that separating the spatial and temporal encoding significantly improves the performance for pose forecasting. To do this, they learned two separate adjacency matrices  $\mathcal{A}_s^{(l)} \in \mathbb{R}^{\mathcal{J} \times \mathcal{J}}$  and  $\mathcal{A}_t^{(l)} \in \mathbb{R}^{\mathcal{T} \times \mathcal{T}}$  for spatial and temporal encoding respectively, instead of a single one  $\mathcal{A}_{st}^{(l)} \in \mathbb{R}^{\mathcal{T}\mathcal{J} \times \mathcal{T}\mathcal{J}}$  (shown in Equation 6).

$$h^{(l+1)} = \Omega(\mathcal{A}_s^{(l)} \mathcal{A}_t^{(l)} h^{(l)} \mathcal{W}_{gcn}^{(l)}) \quad (6)$$

For an encoder with  $L - 1$  layers, the output of the last layer  $h^{(L)}$  is passed through a Temporal Convolutional Network (TCN) decoder to generate a sequence of 3D human poses in  $\mathcal{F}$  consecutive future frames  $\mathbf{X}_{out} = [x_{\mathcal{T}+1}, x_{\mathcal{T}+2}, \dots, x_{\mathcal{T}+\mathcal{F}}]$

#### 3.2.2 Domain Adaptive Adversarial Learning

Firstly, we ran a very simple experiment with the best performing state-of-the-art method, STS-GCN [25] – we trained a human classifier that takes the latent representation  $h^{(L)}$  as input and predicts the human subject. We found that, it can learn to identify the human subject  $\mathcal{H}$  present in a video with very high accuracy (more than 91%). This shows that these HPF models (similar to STS-GCN) may overfit on the human subjects and thus lose the ability to generalize for pose forecasting. In this section, we discuss our proposed approach to side-step this issue by incorporating domain adaptive adversarial learning [34] with the state-of-the-art HPF methods. As a proof-of-concept, here we apply our proposed approach on the currently best performing method STS-GCN.

Here we leverage the gradient reversal layer (GRL) [34] to design Domain Adaptive Adversarial Learning branch (*DAAL-branch*), and augment it with the main STS-GCN framework as shown in Figure 2. The output of the last encoder layer  $h^{(L)}$  is passed through a gradient reversal

| Loss functions         | MPJPE | P-MPJPE | N-MPJPE |
|------------------------|-------|---------|---------|
| MPJPE                  | 67.1  | 51.8    | 64.4    |
| MPJPE + AngLoss [37]   | 79.5  | 60.2    | 77.1    |
| MPJPE + AngLimit [17]  | 74.7  | 57.7    | 71.9    |
| MPJPE + AngOrient [31] | 68.7  | 52.8    | 66.2    |
| OHJM                   | 66.7  | 52.3    | 65.0    |
| MPJPE + WAO            | 66.8  | 51.9    | 64.3    |
| OHJM + WAO             | 66.5  | 51.4    | 64.1    |

Table 1. **Performance of different loss functions.** All angle loss functions do not help in improving performance. Our proposed loss functions are promising. Improved results are highlighted in blue and the best are in bold. (The error is shown in *mm* unit, lower is better)

layer, which – during forward pass – works as an identity function and outputs  $h^{(L)}$  as well. Then a weighted-average of  $h^{(L)}$  is taken along the temporal axis that generates  $h_{adv} = h^{(L)}\mathcal{W}_{adv}^{(L)}$  (where,  $h_{adv} \in \mathbb{R}^{C^{(L)} \times \mathcal{J}}$ ). Here the weights in  $\mathcal{W}_{adv}^{(L)} \in \mathbb{R}^{\mathcal{T}}$  are learned along with the rest of the parameters of the architecture.  $h_{adv}$  is then passed as input of a classifier (i.e., a linear layer followed by a softmax normalization) to generate the probability distribution over all the human subjects in the training set. We then compute the cross-entropy loss considering the ground truth as  $\mathcal{H}$ . The calculated gradient of the loss during the backward pass [23] propagates through the layers in an usual fashion, except for the GRL which reverses the gradients. This generates  $g^{(L)} = -\lambda * g_D$ , where,  $g_D \in \mathbb{R}^{C^{(L)} \times \mathcal{J} \times \mathcal{T}}$  and  $g^{(L)} \in \mathbb{R}^{C^{(L)} \times \mathcal{J} \times \mathcal{T}}$  represents the gradients before and after they propagate through the GRL, and  $\lambda$  is a scalar multiplier to control the amount of adversarial signal (a hyperparameter). This gradient  $g_D$  then propagates through the STS-GCN encoder, along with the gradient of the MPJPE loss from the main branch for generating poses, as shown in Figure 2). Optimizing these two losses using this approach encourages the STS-GCN encoder to generate embedding  $h^{(L)}$  that is harder to classify for the *DAAL-branch*.

## 4. Experiments

We train and evaluate our method on the Human 3.6M dataset [6]. The dataset consists of 3.6 million 3D human poses and corresponding images. There are 32 joints in the skeleton and 15 different actions performed by 7 different actors (i.e., human subjects). The code is implemented with PyTorch framework and is trained on Google Colab

| Loss functions | MPJPE | P-MPJPE | N-MPJPE |
|----------------|-------|---------|---------|
| MPJPE          | 61.2  | 47.3    | 59.1    |
| OHJM           | 61.8  | 47.6    | 59.4    |
| MPJPE + WAO    | 61.2  | 46.8    | 58.6    |
| OHJM + WAO     | 60.5  | 46.8    | 58.1    |

Table 2. **Performance of OHJM and WAO loss after 50 training epochs.** Using both OHJM and WAO reduces the error. Blue numbers are improvements and bold numbers are the lowest errors. (The error is shown in *mm* unit, lower is better)

GPU (P100 and T4). All experiments are evaluated with MPJPE metrics. Experiments on HPE have additional metrics called P-MPJPE and N-MPJPE.

### 4.1. Experiments on Human3.6M with PoseFormer

**Dataset.** From the PoseFormer baseline results, the most challenging actions “Walking dog”, “Sitting down”, and “Photo” are picked. Besides that, the same five subjects (S1, S5, S6, S7, S8) for training and two subjects (S9, S11) for testing are used. The 2D pose detector CPN [3] generates the 2D pose inputs for both training and testing.

**Implementation details.** We use AdamW optimizer [14] with  $\beta_1 = 0.9, \beta_2 = 0.999$ , decayed weight 0.1 and initial learning rate  $4 \times 10^{-5}$ . The learning rate is decreased by 1% every epoch. The PoseFormer is trained in 10 epochs in ablation studies and 50 epochs in the full experiments with a batch size of 512.

#### 4.1.1 Joint angle loss ablation study

Table 1 illustrates the performance of the PoseFormer model after 10 training epochs. Among all surveyed losses, the combination of MPJPE and AngOrient [31] produced approximate errors as the baseline. This AngOrient is used in our next experiment as additional angle constraints.

#### 4.1.2 Online Hard Joint Mining and Wing Angle Orientation

As being mentioned in the Sec. 3.1.3 and Sec. 3.1.2, the MPJPE and AngOrient loss functions face gradient and outlier problems during training. The last section in Table 1 illustrates the results of the proposed loss functions after training for 10 epochs with the baseline PoseFormer. All proposed losses produce lower errors, especially the combination of MPJPE and WAO gives better results in all evaluation protocols.

Table 2 shows the errors collected between using MPJPE loss and our suggested losses after training for 50 epochs. While WAO yields lower errors in all experiments, the OHJM produces a slightly higher error than the baseline. However, when using both OHJM and WAO, the PoseFormer produces the lowest error in all three protocols.

### 4.2. Experiments for DA-STG-GCN

#### 4.2.1 Experimental Setup

**Datasets.** We considered 22 out of the 32 joints similar to previous studies [11, 15, 16, 25].

**Implementation details.** Adam [8] optimizer and a batch size of 512 are used. With the initial value of 0.04, the learning rate is decreased tenfold for every 10 epochs after the 15-th epoch. The training for all our experiments were run for 80 epochs.

| Method           | Walk        | Eat         | Smoke       | Disc.       | Dir.        | Greet.      | Phone        | Pose         | Purch.      | Sit         | SitD.       | Photo       | Wait        | WalkD.       | WalkT.      | Avg.        |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|-------------|-------------|-------------|-------------|-------------|--------------|-------------|-------------|
| ConvSeq2Seq [11] | 82.3        | 87.1        | 81.7        | 129.3       | 115.8       | 147.3       | 114.0        | 187.4        | 151.5       | 120.7       | 150.3       | 128.1       | 117.7       | 162.4        | 87.4        | 124.2       |
| LTD-50-25 [15]   | 60.3        | 75.8        | 72.1        | 118.5       | 105.5       | 136.8       | 105.1        | 174.8        | 134.9       | 118.7       | 143.8       | 118.8       | 108.3       | 146.4        | 65.7        | 112.4       |
| DCT-RNN-GCN [16] | 58.1        | 75.5        | 69.5        | 119.8       | 106.5       | 138.8       | 105.0        | 178.2        | 134.2       | 115.7       | 143.6       | 115.9       | 108.2       | 146.9        | 64.9        | 112.1       |
| STS-GCN [25]     | <i>51.8</i> | <i>52.4</i> | <i>50.0</i> | <i>78.8</i> | <i>71.0</i> | <i>91.6</i> | <b>66.1</b>  | <i>106.4</i> | <i>93.5</i> | <b>75.2</b> | <i>94.3</i> | <i>76.9</i> | <b>72.0</b> | <i>102.6</i> | <i>51.1</i> | 75.6        |
| DA-STS-GCN       | <b>47.8</b> | <b>47.5</b> | <b>46.5</b> | <b>75.8</b> | <b>68.3</b> | <b>90.1</b> | <i>101.9</i> | <b>93.3</b>  | <b>71.1</b> | <i>91.1</i> | <b>71.2</b> | <b>68.1</b> | 99.7        | <b>68.3</b>  | <b>49.3</b> | <b>72.4</b> |

Table 3. **Our DA-STS-GCN outperforms previous models for HPE.** The proposed method yields lower MPJPE errors for most of the actions. The lowest and second lowest errors are highlighted in bold and italic respectively (lower is better).

| Subject IDs |     |      | No. actions |      | $\Delta\lambda$ | MPJPE(mm) |         |
|-------------|-----|------|-------------|------|-----------------|-----------|---------|
| Train       | Val | Test | Train/Val   | Test |                 | Ours      | STS-GCN |
| 1, 6-9      | 11  | 5    | 15          | 15   | 0.004           | 72.39     | 75.15*  |
| 1, 5-7      | 8   | 9,11 | 15          | 15   |                 | 68.07     | 69.22   |
|             |     |      | 10          | 5    |                 | 79.46     | 79.98   |
| 1, 5-8      | 11  | 9    | 10          | 5    | 0.01            | 74.45     | 74.74   |
|             |     |      | 10          | 5    | 0.001           | 97.01     | 97.50   |
| 1, 5-7      | 8   | 9,11 | 5           | 10   |                 | 109.10    | 110.24  |
|             |     |      | 3           | 3    |                 | 164.36    | 269.16  |

Table 4. **Our proposed method outperforms STS-GCN in different configurations.** With different subjects and actions, our model produce lower MPJPE errors than the baseline with a significant margin. Here  $\Delta\lambda$  is the increment rate of  $\lambda$  in our framework (\* the number reported in the paper is 75.6).

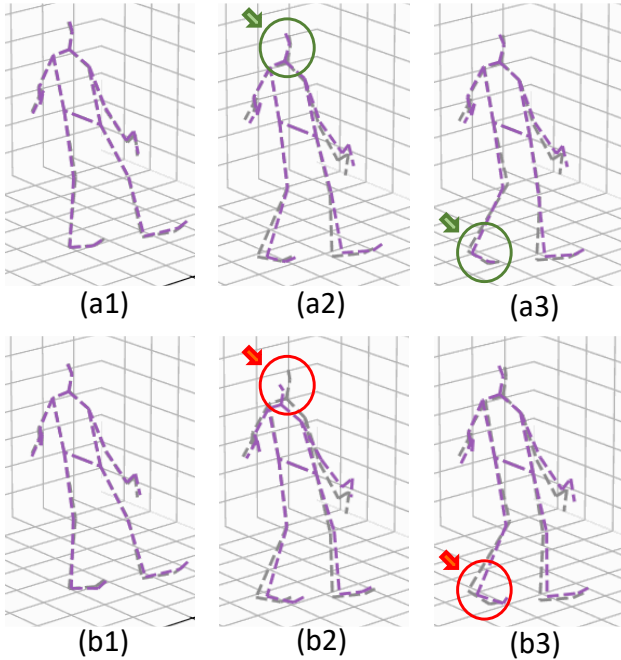


Figure 3. **A comparison between the qualitative results of our method (DA-STS-GCN) and baseline (STS-GCN) for a randomly selected video (activity: walking together).** Here, the first and second row show results from DA-STS-GCN and STS-GCN, respectively. In each row, the figures from left to right represent sequential frames.

#### 4.2.2 Experimental Results for DA-STS-GCN

In Table 3, we depict a comparison of our proposed method DA-STS-GCN with other state-of-the-art methods. Here we can see that DA-STS-GCN outperforms all the previous methods in overall MPJPE (the last column). Moreover, for 12 out of 15 actions, it achieved significantly lower

MPJPE than the next best method and our baseline STS-GCN, while for the rest of the actions it achieved the second best MPJPE.

To evaluate the robustness of our method in different scenarios (varied by different combinations of subjects, actions, and values of  $\Delta\lambda$ ), we conducted an ablation study and show the results in Table 4. The experiment presented in the first row was also reported by [25] to compare their framework STS-GCN (our baseline) with other previous frameworks. In this experiment, our method outperformed STS-GCN (and hence the other methods as well) with a significant margin. Please note that, if we consider the MPJPE reported by [25] (i.e., 75.6), the margin is even larger. In most of the experiments we ran, the MPJPE was lower than that of the baseline regardless of the value of  $\Delta\lambda$ . However, we adjusted  $\Delta\lambda$  separately for each of our experiments to get the best result. In Figure 3, we also provide comparison between the qualitative results of DA-STS-GCN and the baseline, STS-GCN. Here, we can see the forecasted pose from DA-STS-GCN (frames a2 and a3) are more aligned with the ground truth than that of STS-GCN (frames b2 and b3).

## 5. Conclusion and Future Work

In this project, we present some modifications to improve the state-of-the-art models in 3D HPE and HPF. While the OHJM and WAO loss functions are proposed to support PoseFormer in HPE, the DA-STS-GCN leverages robustness over human subjects to improve HPF results. Our preliminary results give empirical evidence that DA-STS-GCN achieves significantly better performance due to increased robustness of the backbone architecture. However, we believe more experimental results are necessary to – (1) provide stronger empirical evidence, (2) understand what are the probable conditions for it to surpass the baseline with a good margin, and (3) evaluate the same pipeline for robustness across various datasets and actions. We are currently working on these experiments, along with extending the current domain adaptation techniques and loss functions for both HPE and HPF with the target of an end-to-end model for robust 3D Pose Estimation and Forecasting in the next two months.

## References

- [1] Bruno Artacho and Andreas Savakis. Unipose: Unified human pose estimation in single images and videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7035–7044, 2020. [1](#)
- [2] Tianlang Chen, Chen Fang, Xiaohui Shen, Yiheng Zhu, Zhili Chen, and Jiebo Luo. Anatomy-aware 3d human pose estimation with bone-based pose decomposition. *IEEE Transactions on Circuits and Systems for Video Technology*, 2021. [1](#)
- [3] Yilun Chen, Zhicheng Wang, Yuxiang Peng, Zhiqiang Zhang, Gang Yu, and Jian Sun. Cascaded pyramid network for multi-person pose estimation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7103–7112, 2018. [4](#)
- [4] Zhenhua Feng, Josef Kittler, Muhammad Awais, P. Huber, and Xiaojun Wu. Wing loss for robust facial landmark localisation with convolutional neural networks. *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2235–2245, 2018. [1](#), [2](#)
- [5] Klaus Greff, Rupesh K Srivastava, Jan Koutník, Bas R Steunebrink, and Jürgen Schmidhuber. Lstm: A search space odyssey. *IEEE transactions on neural networks and learning systems*, 28(10):2222–2232, 2016. [1](#)
- [6] Catalin Ionescu, Dragos Papava, Vlad Olaru, and Cristian Sminchisescu. Human3.6m: Large scale datasets and predictive methods for 3d human sensing in natural environments. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2014. [2](#), [4](#)
- [7] A. Jain, S. Savarese A. R. Zamir, and A. Saxena. Structural rnn: Deep learning on spatio-temporal graphs. *IEEE Computer Vision and Pattern Recognition (CVPR)*, 2016. [1](#), [2](#)
- [8] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014. [4](#)
- [9] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. *The International Conference on Learning Representations (ICLR)*, 2017. [1](#), [3](#)
- [10] Kyoungoh Lee, Inwoong Lee, and Sanghoon Lee. Propagating lstm: 3d pose estimation based on joint interdependency. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 119–135, 2018. [1](#)
- [11] Chen Li, Zhen Zhang, Wee Sun Lee, and Gim Hee Lee. Convolutional sequence to sequence model for human dynamics. *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. [3](#), [4](#), [5](#)
- [12] Hung Yu Ling, Fabio Zinno, Cheng George, and Michiel van de Panne. Character controllers using motion vaes. *International Conference on Computer Vision (ICCV)*, 2021. [1](#)
- [13] Ruixu Liu, Ju Shen, He Wang, Chen Chen, Sen-ching Chung, and Vijayan Asari. Attention mechanism exploits temporal contexts: Real-time 3d human pose reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5064–5073, 2020. [1](#)
- [14] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017. [4](#)
- [15] Wei Mao, Miaomiao Liu, Mathieu Salzmann, and Hongdong Li. Learning trajectory dependencies for human motion prediction. *The IEEE International Conference on Computer Vision (ICCV)*, 2019. [3](#), [4](#), [5](#)
- [16] Wei Mao, Miaomiao Liu, and Mathieu Salzmann. History repeats itself: Human motion prediction via motion attention. *The European Conference on Computer Vision (ECCV)*, 2020. [1](#), [2](#), [3](#), [4](#), [5](#)
- [17] Wei Mao, Miaomiao Liu, and Mathieu Salzmann. Generating smooth pose sequences for diverse human motion prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 13309–13318, 2021. [1](#), [2](#), [4](#)
- [18] Gyeongsik Moon and Kyoung Mu Lee. I2l-meshnet: Image-to-lixel prediction network for accurate 3d human pose and mesh estimation from a single rgb image. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part VII 16*, pages 752–768. Springer, 2020. [1](#)
- [19] Georgios Pavlakos, Xiaowei Zhou, and Kostas Daniilidis. Ordinal depth supervision for 3d human pose estimation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 7307–7316, 2018. [1](#)
- [20] Dario Pavllo, Christoph Feichtenhofer, David Grangier, and Michael Auli. 3d human pose estimation in video with temporal convolutions and semi-supervised training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7753–7762, 2019. [1](#)
- [21] Koutilya PNVR, Hao Zhou, Hugo Larochelle, and David Jacobs. Sharingan: Combining synthetic and real data for unsupervised geometry estimation. *Computer Vision and Pattern Recognition (CVPR)*, 2020. [2](#)
- [22] Davis Rempe, Tolga Birdal, Aaron Hertzmann, Jimei Yang, Srinath Sridhar, and Leonidas J. Guibas. Humor: 3d human motion model for robust pose estimation. *International Conference on Computer Vision (ICCV)*, 2021. [1](#)
- [23] David E Rumelhart, Geoffrey E Hinton, and Ronald J Williams. Learning representations by back-propagating errors. *nature*, 323(6088):533–536, 1986. [4](#)
- [24] Abhinav Shrivastava, Abhinav Kumar Gupta, and Ross B. Girshick. Training region-based object detectors with online hard example mining. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 761–769, 2016. [1](#), [2](#)
- [25] Theodoros Sofianos, Alessio Sampieri, Luca Franco, P. Huber, and Fabio Galasso. Space-time-separable graph convolutional network for pose forecastings. *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021. [1](#), [2](#), [3](#), [4](#), [5](#)
- [26] Alexander Toshev and Christian Szegedy. Deeppose: Human pose estimation via deep neural networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1653–1660, 2014. [1](#)
- [27] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in neural information processing systems*, pages 5998–6008, 2017. [1](#)
- [28] Petar Veličković, Guillem Cucurull, Arantxa Casanova,

- Adriana Romero, Pietro Liò, and Yoshua Bengio. Graph attention networks. In *International Conference on Learning Representations*, 2018. [3](#)
- [29] B. Wang, E. Adeli, H. Chiu, D. Huang, and J. C. Niebles. Imitation learning for human pose prediction. *IEEE International Conference on Computer Vision*, 2019. [1](#), [2](#)
- [30] Bin Xiao, Haiping Wu, and Yichen Wei. Simple baselines for human pose estimation and tracking. In *Proceedings of the European conference on computer vision (ECCV)*, pages 466–481, 2018. [1](#)
- [31] Meng Xiao, Hailun Xia, Ziwei Xie, and Chunyan Feng. Multi-scale recalibration with advanced geometry constraints for 3d human pose estimation. In *2020 IEEE 6th International Conference on Computer and Communications (ICCC)*, pages 1317–1322. IEEE, 2020. [1](#), [2](#), [4](#)
- [32] Liang Yan, Bin Fan, Shiming Xiang, and Chunhong Pan. Adversarial domain adaptation with a domain similarity discriminator for semantic segmentation of urban areas. *2018 25th IEEE International Conference on Image Processing (ICIP)*, pages 1583–1587, 2018. [2](#)
- [33] Sijie Yan, Yuanjun Xiong, and Dahua Lin. Spatial temporal graph convolutional networks for skeleton-based action recognition. In *Thirty-second AAAI conference on artificial intelligence*, 2018. [3](#)
- [34] Evgeniya Ustinova Yaroslav Ganin, Pascal Germain Hana Ajakan, Francois Laviolette Hugo Larochelle, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 2021. [2](#), [3](#)
- [35] Ailing Zeng, Xiao Sun, Fuyang Huang, Minhao Liu, Qiang Xu, and Stephen Lin. Srnet: Improving generalization in 3d human pose estimation with a split-and-recombine approach. In *European Conference on Computer Vision*, pages 507–523. Springer, 2020. [1](#)
- [36] Ce Zheng, Sijie Zhu, Matias Mendieta, Taojiannan Yang, Chen Chen, and Zhengming Ding. 3d human pose estimation with spatial and temporal transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11656–11665, October 2021. [1](#)
- [37] Xiangtao Zheng, Xiumei Chen, and Xiaoqiang Lu. A joint relationship aware neural network for single-image 3d human pose estimation. *IEEE Transactions on Image Processing*, 29:4747–4758, 2020. [1](#), [2](#), [4](#)
- [38] Xiaowei Zhou, Menglong Zhu, Spyridon Leonardos, Konstantinos G Derpanis, and Kostas Daniilidis. Sparseness meets deepness: 3d human pose estimation from monocular video. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4966–4975, 2016. [1](#)